

Messrs.:

Thyssenkrupp Systems Engineering India

MIDC Chakan, Pune,
Maharashtra, India

Subj: Quotation for Pneumatic Manipulators – **Fixed Column Type**
for Mercedes Benz Poland Bodyshop project – **Front Doors**
handling

Dear Sir,

Thank you for your interest shown in Dalmec and we want to
thank you for offering us the opportunity to supply our

Budgetary Offer No. 26136-A Dated: 16th February 2026

We firmly believe that our Pneumatic Manipulators are well suited
to meet the specific requirements and safety of your application
for Bodyshop applications.

The quotation has been finalized purely based on initial details
received and discussions with project team and confirmation on
scope - any change in process further will have impact on
commercials. Feasibility study will be done once complete data /
drawings is received for the project including trolley, fixture,
layout, variants and other data. Any additional accessories if
required, will be at extra cost in buyer scope

Should you need additional information or if any ambiguities arise,
please feel free to contact us at your earliest convenience.

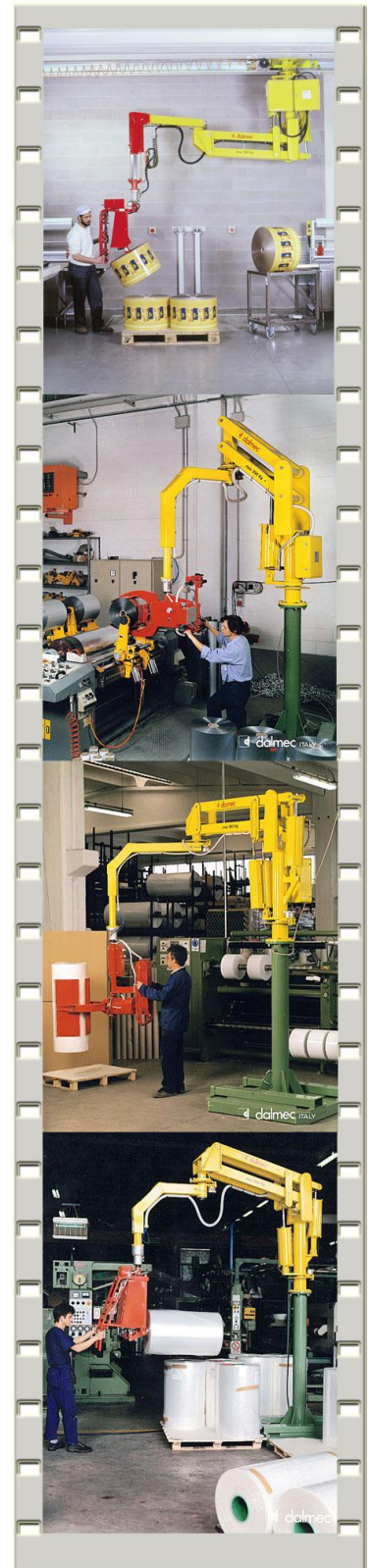
DALMEC looks forward to working with you on this project.

Sourabh Nimodiya

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Manager – Technical Sales

16th February 2026



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Maharashtra, India

OFFER No. 26136-A Dated 16.02.2026

1. Project data

Our internal references for this project are:

Customer number: 06

Your contact persons for this project are:

Sourabh Nimodiya	Tel: +91-7720099724	sourabh.nimodiya@dalmec.in
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2. Project description

Thyssenkrupp India – Mercedes Benz Poland Bodyshop Project – **Fixed Column Type Manipulators for Front Doors handling**

3. Applications

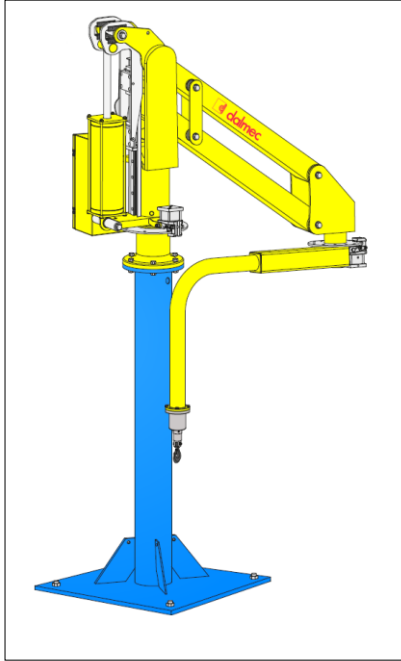
Product	Variants	Max. weight of Product
Front Doors	FA11 – LH and RH FA12 – LH and RH	22.5 kg

Please Note:

- Max. weight capacity of Manipulator: 25kg – Load test / SWL to be done at this weight only
- 3D drawing of all Door variants, input / output trolley, fixture and layout is required to confirm if all variants can be handled with common gripper arrangement. Currently it is considered that all Door variants have common gripping area and common gripper design is feasible. Any change required in input / output trolley and fixture to help commonize the gripper design or addition in accessories in buyer scope

4. Manipulator:

Partner – Fixed Column Type



Pneumatic Manipulator "**PARTNER**" type, column mounted version with base-plate to be fixed to the floor, model with parallelogram and angled terminal arm.

The Manipulator will allow 360° continuous rotation at the units' main axis, 300° rotation between parallelogram and terminal arm, and 360° continuous rotation at the tooling's vertical axis.

The Manipulator will be equipped with a brake on the vertical axis and brakes on all horizontal axes, which will allow the operator to apply the brakes as required per the application's sequence of operation. The brakes are activated by a pushbutton ergonomically positioned on the tooling. When the brakes are applied, the vertical position of the tooling will be fixed and the rotation between the mounting flange and the trunk of the

Manipulator, the terminal arm and parallelogram, and the terminal arm and the tooling will be controlled.

The Manipulator will be equipped with a custom-made gripping tool as below described, specifically designed to handle the respective product in working area

Load balancing is obtained through a pneumatic cylinder actuated by two (2) distinct control circuits:

- The **first** circuit is used to balance the tooling in an unloaded condition.
- The **second** circuit is activated by the operator through the grip pushbutton ergonomically positioned near the operator handgrips. This circuit controls the balancing of the tooling in a loaded condition. When activated the load can be handled in a balanced state through-out the entire work area of the manipulator.

Once the second circuit is activated, the load can be moved in every direction without any further command.

The Manipulator is complete with the following features:

- Cycle counter : to inspect the number of working cycles performed, in order to update periodically the service register in the Instruction Manual.
- Air filter with pressure regulator and automatic condensate outlet

- One control panel with a non-removable lock and two keys, to prevent tampering by unauthorized personnel.
- One adjustable vertical limit bar : once set, the maximum upper and minimum lower vertical heights can be controlled within the work area. The vertical limit is complete with a removable cover plate to prevent tampering by unauthorized personnel.
- One column to be mounted to the customer's floor.

a. Accessories :



Preselector

Allows quick and easy selection of different predictable weights. Control is complete with pointer and large dial for inscription of reference part numbers and/or weights. This requires operator adjustment.



BRAKE SYSTEM that prevents the manipulators' gripping tool from self-drifting around the main vertical rotating axis. The brake system is actuated through a single push-button command mounted on the tooling.



AIR BOOSTER the booster regulator, when connected to air supply line, increases pressure up to two times. The desired pressure increase is easily adjusted in order to feed the Manipulator a compressed air supply min. 0,8 constant Mpa.



DALMEC SAFETY VALVE: A special safety device is automatically inserted if the operator accidentally commands "balancing" without a load or if the operator commands "release" before positioning the load. This device monitors the air flow of the balancing cylinder, dampening the arm's movement if the speed exceeds the set limit.



IN-BUILT AIR RESERVIOR - A unique air reservoir is built into the structure of the Manipulator, if the air supply is suddenly lost, the reservoir will hold sufficient air for the handling cycle to be completed. Upon final consumption of the compress air the unit will gently drift to the ground.

5. Painting

The painting process consists of surface phosphate degreasing, two coats of acrylic water-based primer and dual coats of two components water-based acrylic enamel (tot. min. thickness. 90 μ – approx. resistance capability 200 hours in saline mist – ISO 7253) with our standard colors:

- Body and Arms : Yellow RAL 1023
- Column : Blue RAL 5010
- Base : Blue RAL 5010

Other different colors may be applied upon request at an additional cost of **Euro 500/- per manipulator with additional time**

6. Performance features of Dalmec manipulator

- Simple Operation : the equipment can be used by workers without extensive training.
- Accurate construction and perfect load balancing will offer an ergonomic and comfortable handling system for the operator.
- Any operator, regardless of age and fitness can handle heavy and bulky loads.
- The benefit of a safer working environment reduces risk of accidents.
- A Manipulator reduces fatigue and back injuries, thereby increasing productivity and reducing workers compensation costs.

- Maintenance Free : as long as the Manipulator is supplied with clean dry air, the equipment is virtually fault free.
- Precise movements reduce potential product damage due to rough handling.

7. Safety (for Dalmec Manipulator with Gripper)

The Manipulator will be manufactured in accordance with the essential safety provisions mentioned in chapter 1 and 4 - enclosure I - of the European Standard 06-42 "Directives on Machinery". Among those are :

Directives/Standards	Subject
Directive 2006/42/EC	Machinery directive
Directive 2014/68/EU	Pressure equipment
Standard EN ISO 12100:2010	Safety of machinery. General principles for design. Risk assessment and risk reduction
Standard EN 14238:2009	Cranes. Manually controlled load manipulating devices
Standard EN ISO 13857:2008	Safety of machinery. Safety distances to prevent hazard zones being reached by upper and lower limbs
Standard EN 349:2008	Safety of machinery. Minimum gaps to avoid crushing of parts of the human body
Standard EN 1005-3:2009	Safety of machinery. Human physical performance Part 3: Recommended force limits for machinery operation
Standard EN ISO 4414:2012	Pneumatic fluid power. General rules and safety requirements for systems and their components
Standard EN 14120:2015	Safety of machinery. Guards. General requirements for the design and construction of fixed and movable guards
Standard EN ISO 13850:2015	Safety of machinery. Emergency stop. Principles for design
Standard EN 574:2008	Safety of machinery. Two hand control devices. Functional aspects. Principles for design
Standard EN 842:2009	Safety of machinery. Visual danger signals. General requirements, design and testing
Standard EN 1037:2008	Safety of machinery. Prevention of unexpected start up
Standard EN 614-2:2009	Safety of machinery. Ergonomic design principles Part 2: Interactions between the design of machinery and work tasks
Standard ISO 8573-1:2010	Compressed air. Contaminants and purity classes

Several safety features are designed in all Dalmec manipulators; the following ones are the main ones to be integrated in the above described unit :

SAFETY FEATURES of Dalmec Manipulators:

- A Unique Air Reservoir is built into the structure of the Manipulator, if the air supply is suddenly lost, the reservoir will hold sufficient air for the handling cycle to be completed. Upon final consumption of the compress air the unit will gently drift to the ground.
- All Dalmec units include Pneumatically Actuated Brakes to effectively hold the main body avoiding sideway and upward drift motions of the unit when the Manipulator is not in use. All brakes are simply activated through a pneumatic push button. All brakes are spring loaded, and will be actuated automatically in case of insufficient compress air supply. These brakes do not BLOCK but only “stiffen” the movements of the Manipulator.
- The main axis and intermediate axis rotation span is adjustable; this to limit the angular movements of the arm in case of obstructions in the working area.

8. Documentation

An Instruction Manual will be supplied in English language, in compliance with the European Standard 06-42 "EC Directives on Machinery", containing the following chapters:

- general instructions and regulations
- description and technical data
- transportation and installation
- tuning and operating instructions
- safety devices
- maintenance
- trouble shooting
- pneumatic diagram

Other documentation may be provided upon extra charge.

The documentation does not include the supply of constructional drawings, being these strictly confidential and sole properties of Dalmec.

The Manipulator will be complete with CE marking, in accordance with chapter 1.7.4 – encl. I of the European Standard 98-37 : “Directives on Machinery”.

The EC Certificate of Conformity, in compliance with enclosure II of the above mentioned Standard, will be supplied together with the machine.

9. Installation, final tuning and operator training

Installation of the Manipulator at customer plant will be performed by our fully trained service engineer (1no.); it is required that your personnel will assist our engineers during all installation phases. One of our engineer will supervise and instruct your personnel. The complete scope of equipments, structure, manpower others as requested by Dalmec technician for Manipulator installation will be in buyer scope. 1no. Dalmec engineer will be present at site for supervision of installation

Term of installation:

- Duration of Installation : 4 working days per manipulator including trials and training. Addition support if required will be at extra cost in buyer scope
- One working day is to be considered 8 working hours.
- Food and lodging arrangements will be at Dalmec cost

At design sign-off, Dalmec will provide the following documents :

- Total weight of manipulator
- Load values on anchors.

It is customer's duty to check the compatibility of the above data with the factory construction standards; Dalmec will proceed with the installation only after receiving a written confirmation that the above check results positive.

10. Exclusions

The below mentioned items are not included in the scope of our supply and are at your charge:

- Overhead structure, Site preparation prior to Manipulator installation (floor reinforcement, cleaning, etc.)
- Site preparation prior to Manipulator installation (floor reinforcement, cleaning, etc.)
- Offloading the manipulator and related accessories when they are delivered to your site.
- Provision of lifting equipment for installation and handling all material, including driver service. (e.g. Forklift truck, scissor platform, etc.) The list of needed equipment will be specified by Dalmec engineers at least one week before the installation will start.
- Construction and installation of steel structure to hold the overhead rail/manipulator. (Dalmec will provide all needed load schemes at project DAP sign off)
- Connection of the compressed air supply to the attachment point of the Manipulator.
- Integration with PLC, electrical components, others if required will be in Customer scope with any accessories required for integration
- Any spares required for alongwith system

- Festooning, Cabling and Routing of Electric items if required, limit Switches, Home position / other sensors and mounting brackets required for the same will be in Customer scope. Dalmec will provide guidance for mounting on manipulator for additional Switches/Sensors
- Any additional charges due to delay in material unloading/demurage/others
- Any other scope which is not mentioned in this offer

11. Warranty

- Dalmec warrants all products supplied by them to be free from manufacturing defects in materials and workmanship for a period of twelve (12) months.
- The start of warranty period will coincide with signing of the acceptance report. In case the machine is not accepted but is still used for production purposes, the start of warranty period will coincide with the date of production start.
- In case none of the above conditions will be met, warranty will in any case start 3 month after delivery of the manipulator.
- Dalmec guarantees the ultimate quality of all materials and manufacturing accuracy along with correct operation.
- Components of normal wear and tear are not included in the above warranty.
- Equipment damage resulting from incorrect use, missed respect to all instructions of operation and care outlined in the Instruction Manual are to be excluded.
- Subsequent delivery of spare parts and parts subject to wear is guaranteed for 10 years after delivery.

All parts proven to be manufacturing defective during the warranty period will be replaced and/or repaired at no charge under the following conditions :

- prompt delivery
- delivery DDP (Incoterm – 2000) Dalmec Pune.
- packing included

All on-site visits of our technician needed for warranty related issues will have to be agreed in advance with our service department.

The warranty does not cover, for any reason, liabilities related to the inefficiency period of the Equipment.

12. Samples

A representative sample of products to be handled is required by Dalmec; the samples will be used to conduct handling tests during the final manufacturing phases of the manipulator.

The following conditions will apply :

- The samples will be delivered DDP (Incoterm – 2000) Dalmec SpA, Cles (Italy).
- The samples should preferentially be defective parts with low or no commercial value, but still representative of the items to be handled.
- The samples will be returned to your freight forwarder after trials.

13. Design approval

Before starting production of the manipulator, a complete set of engineering 2D drawings for unit will be provided to your engineers for approval. Dalmec will not manufacture any machine without written approval from your engineers.

The drawings to be approved are :

- Manipulator overall assembly drawing in Dalmec standard format
- Tooling overall assembly drawing in Dalmec standard format
- Load scheme for manipulator erection (forces on floor rods, torque on overhead flange, weights, etc.)

If structural changes or severe modifications are required after written approval, all related costs will be at your own charge.

14. Confidentiality

This offer is strictly confidential.

The offer, or part of it, cannot be disclosed to third party without a written consent from Dalmec.

➤ **Project Scope:**

15. Front Doors Handling Manipulator

15.1 Tooling Similar Photo – Front Doors handling:



The gripping tool, connected with the Manipulator through a pneumatic rotating joint, is composed of gripper arrangement specially designed to handle the Front Doors in Bodyshop area from Fixture to Trolley.

Gripping and automatic pre-set load balancing are obtained automatically simply by activating two pneumatic push buttons ergonomically positioned on the tooling near operators hand grips.

Release and automatic no-load balancing are obtained, simply by operating a pneumatic push-button ergonomically mounted on the tooling near operators handgrips.

Application is pick and place type only – no inclination required in product during handling i.e. only horizontal to horizontal handling of Front Doors is required with manipulator

15.2 Operating Procedure – Front Door Handling:

1. Pick Manipulator from parking position and move it towards the Door Fixture on which Door is kept in Horizontal condition with its inner side facing sky
2. Locate Gripper onto inner surface of Door for clamping
3. Press Grip button on tooling handle to clamp the Door with Gripper and check if positive gripping indicator has turned green
4. Arm and lift the Door, handling in balanced condition move it towards Output trolley
5. Locate the Door onto Trolley and release
6. Move the manipulator home position for next cycle

15.3 Technical Specs – Front Door handling Manipulator:

Manipulator Type:	Partner – Fixed Column Type
Working range refer to product:	Approx.. Max. 2500 mm – as per final layout reach will be finalized for respective station
Vertical lift:	min. 800 mm
	max. 1200 mm
Max weight capacity:	25kg
Required air pressure:	0.85 MPa constant (clean, dry, oil free air) – Line Pressure of 0.5 MPa required
Weight of the manipulator:	approx. 650 kg
Base Plate Dimension:	approx. 800mm x 800mm

Please Note:

- Overhead structure / Floor requirement and clear space as required for Manipulator in Buyer scope.
- Any changes required in input trolley/dunnage/offline fixture/conveyor fixture, layout, shifting of utilities as per Manipulator and Gripper in Buyer scope
- Only 2 variants of Front Doors needs to be handled with Manipulator and both these variants have common gripping area as confirmed during discussion, if any specific gripping area to be considered and if it is different for each variant it will have an impact on cost
- **Fixed Column Type Manipulator is considered for all stations**
- The gripping area and concept can be finalized once we receive complete 3D drawings of product, trolley, dunnage, conveyor and offline fixture.
- Enough space must be available at all points to open and close Gripping jaws.
- Manipulator position and reach can be finalized during layouting stage after PO receipt.
- There are other equipment near the station which need to remove/offset the equipments as per manipulator requirement – Buyer scope – clear working space must be available for manipulator movement

16. Prices :
16.1 Front Doors handling Manipulator – 25kg weight Capacity:

Description	Unit Basic Price	Qty	Total Basic Price
Front Doors handling Manipulator with Gripper (as described in Chapter-3 and 14)	Euro 65,000/-	06	Euro 390,000/-
Accessories as below:			
- Pre-Selector or Up/Down Lever Control (1no. per manipulator)			
- Brake System (1no. per manipulator)			
- Positive Grip Indicator (1no. per manipulator)			
- Dalmec Safety Valve (1no. per manipulator)			
- Air Booster (1no. per manipulator)			
Total Basic Price for Front Door handling Manipulators – 06 units including above accessory items – Ex-works Italy Basis (HSN: 84289090):			Euro 390,000/-
Road/Sea Freight from Ex-works Italy till Thyssenkrupp Germany Plant - Custom Duty/clearance, VAT, demurrage, other taxes till customer plant in buyer scope			Buyer scope

16.2 Service Scope – Manipulator Installation & Commissioning:

Description	Qty	Total Basic Price
Installation and Commissioning of Manipulator at Mercedes Benz Poland Plant for Total of 4 days duration including trials and training per manipulator set. All required manpower, documentations, others as required for installation and commissioning to be arranged within this time. Additional support if required will be at extra cost of Euro 2000 per day	06 sets	Euro 42,000
Electrical connections to PLC, other electrical connections, Cables, Shifting of utilities, Festooning, cabling/routing, local certifications/other approvals if any	06 sets	Buyer scope
Freight, Custom Clearance, Duties, taxes, VAT, demurrage, others	06 sets	Buyer scope

Please Note:

- For the value in Euro, Purchase Order should be in favor of **“Dalmec SpA, Italy”**.
- Taxes extra as applicable in buyer scope
- **Manipulator HSN Code: 84289090**

- Manipulator Design, Engineering, Manufacturing & Pre-Dispatch Testing/Trials will be done at Dalmec SpA, Italy Plant
- Design Approval (DAP) and complete project coordination will be done with Dalmec India Technical Manager and Key Account Manager for the project
- Overhead Structure/Floor requirements, shifting of utilities, any modifications, others as required at site for Manipulator installation and proving in Buyer scope.
- Any changes in parameter, increase in weight, reach, variants, others will be at additional cost in buyer scope
- Any spares if required will be at additional cost in buyer scope
- Fixed Column Type manipulator is considered for all locations
- Supply Air must be clean and Dry. Air Dryer if required in buyer scope
- Pneumatic connection must be made available at Manipulator input
- No. of Product Variants: 2Nos. with common gripping area
- Local Taxes, Duties, Custom clearance, VAT extra as applicable from Ex-works Italy till customer plant in buyer scope
- Unloading of system at Mercedes Benz Poland plant in buyer scope
- All Manipulator Drawings will be shared in 2D Autocad format with different views and Gripper Drawing will be shared in 3D e-design format
- Additional stand-by support if required will be at extra cost in buyer scope. If local Quarantine is required it will be at extra cost
- In above service cost it is considered that work will be completed in single visit, if multiple visits are required it will be at extra cost
- Integration with PLC will be in Customer scope
- If any specific make list has to be followed for bought out items it will be at extra cost in Buyer scope. Dalmec supplies manipulators to Mercedes Benz Globally hence the entire equipment is approved however if any specific make items has to be considered - cost and time required for procurement will be extra as applicable
- The Maximum weight capacity for Front Door handling system is 25kg and no additional weight even for testing purpose can be considered. All Dalmec systems as designed as per most severe safety and ergonomic standards and provided with CE certificate for given max. weight capacity which in case of Front Door handling application is 25kg
- Any changes required in Conveyor fixture as per Manipulator Gripper in Buyer scope – enough space must be available for manipulator gripper to enter and release the Tray onto conveyor fixture
- Any changes at site, shifting of utilities, modification in structure, others in buyer scope

- Enough space must be available for manipulator movement within the station.
Any changes in layout in buyer scope
- Only CE certification will be provided with Manipulator additional certifications like UL/PE or other if required will be in buyer scope
- Clear site must be available as per mutually agreed schedule to start the installation activity

17. Terms of sale

Delivery period: 24 weeks – **Ex-works Dalmec Italy** from the **date of complete technical data/drawings and receipt of PO and advance.** (All requested Data/Drawings must be shared within 7 days of receipt of PO). Any delay in getting the requested data / drawings will have an impact on manipulator readiness

Delivery Terms: Ex-works Dalmec Italy

Warranty: 1 year. Please refer chapter 9 for more details

Payment Terms: Equipment: 40% advance with PO against Proforma invoice
60% against readiness before dispatch

Payment Terms: Service: 50% advance before installation
50% against installation within 15 days

Order Cancellation: 0% charge Within 30 days of PO receipt
10% after DAP & below 60 days from PO date
In the event of cancelation after 60 days or DAP drawings are approved 100% charges will be applicable

Packing: Included.

The above mentioned manipulators and related accessories will be packaged in a wooden crate/box suitable for Sea/Road transportation.

OFFER VALIDITY = 60 days

Yours faithfully,

Sourabh Nimodiya
Manager – Technical Sales
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